

Introduction to Deep Hedging

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Motivation

- ▶ As market makers, option sellers typically earn a steady stream of small premiums but remain **exposed to rare yet potentially significant losses** caused by extreme market movements (*the option seller's problem*). To manage this risk, effective **hedging strategies** are required.
- ▶ Traditional methods only achieve **perfect hedging** in **complete markets**, under often unrealistic assumptions in practice (continuous trading, constant volatility, perfect liquidity, and frictionless markets). In realistic **incomplete markets**, they offer only **imperfect hedging**.

Deep Hedging

Deep Hedging uses machine learning to learn hedging strategies from market data, accounting for transaction costs, market frictions, past hedging decisions, and other real-world constraints. It performs robustly in **both complete and incomplete markets**, without relying on restrictive theoretical assumptions.

Derivatives

- ▶ A **derivative** is a financial contract whose value is derived from the price of an underlying asset.
- ▶ Derivatives are widely used for:
 - ▶ **Hedging**: reducing exposure to financial risks,
 - ▶ **Speculation**: taking positions on future price movements,
 - ▶ **Risk management**: managing exposures arising from business activities.

Main Types of Derivatives

- ▶ **Forward:** An agreement to buy or sell the underlying at a predetermined price on a specified future date.
- ▶ **Futures:** Similar to a forward contract, but standardized and exchange-traded.
- ▶ **Option:** A contract that gives the holder the right, but not the obligation, to buy or sell an underlying asset at a predetermined price before or at expiration of the contract.

Options

- ▶ A **Call Option** is the right to buy the underlying at a strike price K on (or before) expiration at T .

$$\text{Payoff} = \max(S_T - K, 0) \quad (1)$$

- ▶ A **Put Option** is the right to sell the underlying at a strike price K on (or before) expiration at T .

$$\text{Payoff} = \max(K - S_T, 0) \quad (2)$$

Moneyness

- ▶ **In-the-Money (ITM)**: Price above strike for a call, below strike for a put.
- ▶ **At-the-Money (ATM)**: Price at the strike.
- ▶ **Out-of-the-Money (OTM)**: Price below strike for a call, above strike for a put.

Exercise Style

Options can be classified by when and how they are exercised, which affects the **complexity of their pricing and hedging**.

- ▶ **European Style:** Exercisable only at expiry T . Simpler to hedge, and most research is done on this type of option.
- ▶ **American:** Exercisable at any time up to T . More complex to hedge
- ▶ **Other Exotic Options:**
 - ▶ **Barrier:** Activates if the underlying hits a preset level.
 - ▶ **Asian:** Payoff depends on average price.
 - ▶ **Lookback:** Payoff depends on the best/worst price over its life.
 - ▶ **Basket:** Based on a portfolio of assets.

Greeks

The **Greeks** measure the sensitivity of an option's price to change in its underlying risk factors, using the **partial derivatives** of the price relative to each parameter:

- ▶ **Delta (Δ)**: Sensitivity of the option price to changes in the underlying asset price.
- ▶ **Gamma (Γ)**: Sensitivity of Delta to changes in the underlying asset price.
- ▶ **Theta (Θ)**: Sensitivity of the option price to the passage of time (time decay).
- ▶ **Vega (ν)**: Sensitivity of the option price to changes in market volatility.
- ▶ **Rho (ρ)**: Sensitivity of the option price to changes in interest rates.

*Remark: The Deep hedging approach is generally considered **Greek-free**.*

Hedging

Definition: Hedging is a financial practice where one or more assets are traded to **minimize risk** associated with a correlated financial exposure or liability.

Objective: Reduce the variability of future profits and losses (P&L) caused by **adverse market movements**.

- ▶ Selling an option creates a stochastic **future liability** tied to the underlying asset's price.
- ▶ We want to construct a tracking portfolio that offsets this risk.
- ▶ Because perfect replication/risk elimination is rarely achievable in practice, we require advanced solutions like **Deep Hedging**.

Types of Hedging

- ▶ **Static Hedging:** The hedging portfolio is established when the liability is created and remains unchanged until maturity of the liability.
- ▶ **Dynamic Hedging:** The hedging portfolio is periodically rebalanced as market conditions and risk exposure changes until maturity of the liability.
- ▶ **Continuous Hedging:** A specific type of dynamic hedging where the hedging portfolio is continuously adjusted.

The continuous hedging assumption is commonly used in financial theory but is **unrealistic in practice** due to non-zero transaction costs, discrete trading, market friction, etc.

Remark: Deep Hedging focuses on learning optimal dynamic hedging strategies under realistic market constraints.

Arbitrage-Free Principle

Definition: A market is arbitrage-free if **no strategy can produce risk-free profits**, with zero initial investment.

Remark: Deep hedging does not require that the market is arbitrage-free, unlike other hedging frameworks like Black-Scholes where it is a fundamental assumption. Deep hedging automatically exploits arbitrage opportunities to minimize portfolio risk.

Super-Replication

Definition: Super-replication is a conservative approach to hedging which can be used by a seller of the contingent claim Φ to **remove all residual hedging risk**.

A super-replication strategy for the contingent claim Φ is defined by a pair (v, δ) such that $v \in \mathbb{R}$, and δ is an admissible hedging strategy for which $V_0^\delta = v$ and $V_N^\delta = B_N(v + G_N^\delta) \geq \Phi(S_N, Z_N)$ almost surely with respect to the probability measure $\mathbb{P}$

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¹Carbonneau, A. and Godin, F. *Equal Risk Pricing of Derivatives with Deep Hedging*.

Super-Replication

Let $\bar{v}$ be the greatest lower bound of the set of initial portfolio values for which a **super-replication strategy** exists:

$$\bar{v} := \inf \left\{ v : \exists \delta \in \Pi \text{ such that } \mathbb{P} \left[B_N(v + G_N^\delta) \geq \Phi(S_N, Z_N) \right] = 1 \right\}. \quad (3)$$

$\bar{v}$ is called the super-replication price of Φ , and it represents an upper bound of the set of **arbitrage-free prices** for Φ .²

²Carbonneau, A. and Godin, F. *Equal Risk Pricing of Derivatives with Deep Hedging*.

Sub-Replication

Definition: A sub-replication strategy is a pair (v, δ) that completely **removes the hedging risk exposure** associated with a long position in Φ . (i.e for which $B_N(v + G_N^\delta) \leq \Phi(S_N, Z_N)$) almost surely with respect to the probability measure $\mathbb{P}$)³

³Carbonneau, A. and Godin, F. *Equal Risk Pricing of Derivatives with Deep Hedging*.

Sub-Replication

The **greatest lower bound** of the set of portfolios such this strategy exists is called the **sub-replication price** of Φ and is a lower bound of the set of **arbitrage-free prices** for Φ :

$$\underline{v} := \sup \left\{ v : \exists \delta \in \Pi \text{ such that } \mathbb{P} \left[B_N(v + G_N^\delta) \leq \Phi(S_N, Z_N) \right] = 1 \right\}. \quad (4)$$

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⁴Carbonneau, A. and Godin, F. *Equal Risk Pricing of Derivatives with Deep Hedging*.

Arbitrage-Free Pricing

A price of a contingent claim Φ is **arbitrage-free** if it falls within the interval $[\bar{v}, \underline{v}]$

Complete vs Incomplete Markets

Under a **complete market**, all contingent claims can be perfectly replicated by a self-financing trading strategy using the available asset. This also means that every payoff is attainable.

Example: Black-Scholes Model

A market is an **incomplete market** if there exists some payoff that cannot be perfectly replicated by any admissible strategy. Real markets are incomplete.

Real markets are generally incomplete due to:

- ▶ Discrete-time trading
- ▶ Transaction costs
- ▶ Stochastic volatility
- ▶ etc.

Self-Financing Process

Definition: A self-financing process δ implies the **absence of cash infusions into or withdrawals from** the portfolio except at t_0 .

The process δ is a self-financing trading strategy if it is predictable and if:

$$\delta_{0_{n+1}}^{(1:D)} \cdot S_n + \delta_{0_{n+1}}^{(0)} B_n = \delta_{0_n}^{(1:D)} \cdot S_n + \delta_{0_n}^{(0)} B_n, \quad n = 0, 1, \dots, N-1. \quad (5)$$

Where $\delta_{0_n}^{(1:D)}$ is a vector of the positions in the D risky assets, $\delta_{0_n}^{(0)}$ is the position in the risk-free asset, S_n are the prices of risky asset D at time $t_n \in T$ and B_n the annualized compounded risk-free asset price at time t_n .

Black-Scholes

Definition: The Black-Scholes formula provides an estimate for the **fair price of a call or put option**. The formula assumes lognormally distributed returns and assumes geometric Brownian motion.

We have the following equation for the Black-Scholes formula for a call option:

$$C = S_0 N(d_1) - Ke^{-rT} N(d_2), \quad (6)$$

$$d_1 = \frac{\ln(S_0/K) + \left(r + \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \quad (7)$$

Where $N(x)$ is the CDF of the standard normal distribution.

Delta Hedging

Definition: Delta hedging is a strategy that aims to **neutralize risk associated with the price change of the underlying** by constructing a portfolio to have a total net delta of zero (i.e., $\Delta^{(p)} = 0$).

$$\Delta^{(p)} = \sum_{j=1}^J \theta^{(j)} \Delta^{(j)} \quad (8)$$

where $\Delta^{(j)}$ is the j th asset sensitivity to price change, and $\theta^{(j)}$ the weight of each asset. To find a Delta-neutral portfolio, solving for the weights using this equality $\Delta^{(p)} = 0$

Delta Hedging Drawbacks

Delta-Gamma hedging has a similar approach, neutralizing both the $\Delta^{(p)}$ and $\Gamma^{(p)}$, which is the rate of change of $\Delta^{(p)}$.

This risk management strategy has some drawbacks:

- ▶ Achieving a Delta-neutral position may require frequent adjustments, leading to **high transaction costs**.
- ▶ In a high-volatility market, keeping a delta-neutral position might be challenging.
- ▶ In an incomplete market, it does not create a perfect hedging strategy, meaning there is **some residual risk**.

GBM for the model of S_t

Under Geometric Brownian motion, the stock price is assumed to follow a stochastic differential equation (SDE):

$$dS_t = \mu S_t dt + \sigma S_t dW_t \quad (9)$$

The first term is deterministic and represents the **drift of the stock price** over a small period.

The second term is random and represents the **uncertainty in the stock price** due to the market volatility of the underlying, with Brownian motion increment $dW_t \sim N(0, dt)$

Risk Measure

Definition: A risk measure is a function $\rho(X)$, that assigns a **real number to** X , which is a random payoff. It quantifies the risk the portfolio or position assumes. Generally, a **higher value for a risk measure implies higher risk**.

In an incomplete market, we assume that we **cannot fully hedge all risks**; hence, we want to minimize our risk exposure by minimizing this measure.

Risk Measures properties

The following are coherence axioms for a risk measure applied to liabilities X and Y (i.e., desirable properties for the chosen $\rho(X)$):

- ▶ **Monotonicity:** if $X \leq Y \implies \rho(X) \leq \rho(Y)$, a larger liability is riskier.
- ▶ **Translation invariance:** $\rho(X + m) = \rho(X) + m$ for any constant m , increasing your liability X increases the loss
- ▶ **Positive homogeneity:** $\rho(\lambda X) = \lambda \rho(X)$ for any scaling constant λ , scaling the position scales the risks.
- ▶ **Subadditivity:** $\rho(X + Y) \leq \rho(X) + \rho(Y)$, Diversification does not increase risk.

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⁵Carbonneau, A. and Godin, F. *Equal Risk Pricing of Derivatives with Deep Hedging*.

Convex Risk Measures

A possible choice for a risk measure is the **variance**. When evaluating profits and losses, we often consider only losses to be risky, but variance treats **any movement from the mean as risky**.

Hence, we consider convex risk measures to characterize the hedger's **risk aversion**. We then replace positive homogeneity by the following:

Convexity: for $0 \leq \lambda \leq 1$,
$$\rho(\lambda X + (1 - \lambda)Y) \leq \lambda \rho(X) + (1 - \lambda)\rho(Y)$$

Which simply translates to **diversification not increasing risk**.⁶

⁶Carbonneau, A. and Godin, F. *Equal Risk Pricing of Derivatives with Deep Hedging*.

Value-at-Risk

Definition: Value-at-Risk is a risk measure that quantifies a **loss threshold** over a given time horizon and significance level α . Let X represent the loss of a portfolio, then:

$$VaR_{\alpha}(X) = \inf(x \in \mathbb{R} : P(x \leq x) \geq \alpha) \quad (10)$$

Definition: Conditional Value-at-Risk is a risk measure that measures the average loss in the **tail** beyond the VaR threshold.

$$CVaR_{\alpha}(X) = E(X \mid X \geq VaR_{\alpha}(X)) \quad (11)$$

Quadratic Penalty

Definition: Quadratic Penalty is a risk measure similar to variance, using any target value c as the relative distance instead of the mean. Let $R = -P_x$, the negative profits at time t under strategy x , and set $c = 0$.

$$\rho(R) = E[(R - c)^2] = E[R^2] = E[P_x^2] \quad (12)$$

Definition: The root semi-quadratic penalty eliminates the upside penalty to allow profit taking. It is a **convex risk measure**.

$$\rho^{RSPQ}(R) = \sqrt{E[(R - c)_+^2]} = \sqrt{E[R_+^2]} = \sqrt{E[(-P_x)_+^2]} \quad (13)$$

Volatility Clustering and Fat Tails

The Black-Scholes model relies on strict assumptions that are rarely satisfied by real markets. It fails because of the following:

- ▶ **Volatility Clustering:** Large price moves tend to be followed by other large price moves, whilst small price moves tend to be followed by small price moves. Current volatility depends on past volatility
- ▶ **Fat Tails:** Extreme gains and losses appear more often than what is predicted by a normal distribution
- ▶ **Leverage Effect:** Negative returns are often associated with a later increase in volatility

Deep Hedging can incorporate these market behaviors when they are captured by the training data. It is a robust strategy in real markets.

GARCH Model

Definition: A GARCH(p,q) (Generalized Auto-regressive Conditional Heteroskedasticity) model uses past conditional volatility as input for current conditional volatility. It takes into account volatility clustering.

$$\sigma_t^2 = \omega + \sum_{i=1}^q \alpha_t \varepsilon_{t-i}^2 + \sum_{j=1}^p \beta_t \sigma_{t-j}^2 \quad (14)$$

In practice, GARCH(1,1) is commonly used because of its parsimony, and good empirical performance.

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (15)$$

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⁷Bollerslev, T. *Glossary to ARCH (GARCH)*. Duke University, CREATES, and NBER. Version: February 16, 2009.

GJR-GARCH

Definition: The GJR-GARCH model allows volatility to respond differently to positive or negative innovations (i.e., takes into account the leverage effect) as well as volatility clustering. The GJR-GARCH(1,1) is represented by:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \mathbb{I}(\varepsilon_{t-1} < 0) + \beta \sigma_{t-1}^2, \quad (16)$$

or

$$y_n = \mu + \sigma_n \varepsilon_n, \quad \sigma_{n+1}^2 = \omega + \alpha \sigma_n^2 (|\varepsilon_n| - \gamma \varepsilon_n)^2 + \beta \sigma_n^2, \quad (17)$$

where the model parameters are positive real numbers, and $\mu \in \mathbb{R}$, $\gamma \in \mathbb{R}$, $\{\varepsilon_n\}_{n=1}^N$ is a sequence of independent standard Normal random variables.

GJR-GARCH

Volatility clustering implies that large price moves tend to be followed by other large price moves.

GJR-GARCH takes this into account by making σ_t depend on past shocks ϵ_{t-1}^2 and past volatility σ_{t-1}^2 .

The **leverage effect** refers to the tendency for negative returns to increase future volatility more than positive returns of the same magnitude, due to an increase in the Debt-to-equity ratio when share prices decline.

GJR-GARCH fixes the asymmetry with the term $\gamma \epsilon_{t-1}^2 \mathbb{I}(\epsilon_{t-1} < 0)$, which only activates if the previous shock was negative, and where γ is the asymmetry coefficient.

Gaussian Regime-Switching

The daily log-returns $y_n := \log(S_n/S_{n-1})$ for $n = 1, \dots, N$ are often assumed to follow a **Gaussian regime-switching model**, which implies that the log-returns depend on an unobservable discrete-time process defined by a finite-state Markov Chain.

$$y_n \mid X_n = k \sim N(\mu_k, \sigma_k^2), \quad k = 1, \dots, K \quad (18)$$

Where X_n is the **unobservable regime** at time n . Define $\{X_n\}$ as a discrete Markov chain on $\{1, \dots, K\}$ with transition matrix

$$P = (p_{ij}), \quad p_{ij} = \mathbb{P}(X_{n+1} = j \mid X_n = i) \quad (19)$$

This is a hidden Markov Chain. In short, it estimates the parameters $\theta = \{\mu_k, \sigma_k^2\}_{k=1}^K \cup \{p_{ij}\}$ corresponding, for example, to a "calm" or "crisis" regime.

Reinforcement Learning Foundation

Definition: Reinforcement learning is learning what to do, how to map situations to actions, to **maximize numerical reward**. The agent must discover the best actions by trying them and seeing which provides **higher rewards**.

Any actions taken at a given step of a multi-step process affect current and subsequent rewards. Some rewards are delayed.

Sequential decision problems must involve:

- ▶ A set of possible states of the environment
- ▶ A set of possible actions
- ▶ An objective (rewards that depend on the previous states and actions)

Exploitation vs Exploration tradeoff

The agent must strike a balance of two competing actions to achieve a successful outcome:

- ▶ **Exploiting:** The agent applies actions that were identified as successful, hence exploiting current knowledge.
- ▶ **Exploration:** The agent applies new actions never attempted before to be able to identify more successful actions, hence exploring new situations.

Only taking exploitative actions risks staying in a suboptimal policy, while only exploring does not capitalize on what has been learned before. A balance of both is needed.

Epsilon-greedy approach: To balance both action types, at each step and where $\varepsilon \in [0, 1]$ is the exploration rate, the agent chooses:

$$a_t = \begin{cases} \arg \max_a Q(s_t, a) & \text{with probability } 1 - \varepsilon, \\ \text{a random action} & \text{with probability } \varepsilon, \end{cases} \quad (20)$$

Policy

A **policy** is a mapping from states of the environment to actions.
It may be deterministic or stochastic.

Reward Function

A **reward function** maps states and actions to a numerical amount called the reward that the agent seeks to maximize. It may be deterministic or stochastic.

Value Function

The **value function** estimates the expected cumulative future reward obtained from a given state. Estimating the value function is one of the central problems of reinforcement learning.

Temporal Differences

A common **temporal differences** update rule is:

$$V(S_t) \leftarrow V(S_t) + \alpha [R_{t+1} + V(S_{t+1}) - V(S_t)] \quad (21)$$

Where S_t, S_{t+1} are the current and next states respectively, and α the step-size parameter controlling the rate of learning.

If S_{t+1} has a high probability of "winning" from S_t , then S_t should also be associated with a **high probability** of "winning".

By combining **greedy moves** with some **exploration**, the value estimates eventually converge to the optimal policy.

Learning types

Value-Based: Tries to estimate the value function of the states and tries to optimize it

Policy-Based: Tries to learn the optimal policy without using the value function and directly estimates the policy.

Actor-Critic: Combines both the Value and Policy learning approaches.

Model Environment

A **Model of the environment** predicts the next states and reward given the current state and action.

- ▶ **Model-Based:** The agent has access to the environment's dynamics and can use it to plan its next moves.
- ▶ **Model-Free:** The agent learns directly from experience, without explicitly modeling transitions or rewards.

Bellman Equation

Definition: The **value function** $V_\pi(s)$ gives the expected discounted return starting from state s , discount factor γ and under policy π .

It satisfies the **Bellman equation**, which takes into account the immediate reward plus discounted future value:

$$\begin{aligned} V_\pi(s) &:= \mathbb{E}_\pi [R(s, a) + \gamma V_\pi(s') \mid s] . \\ &= \sum_a \pi(a \mid s) \sum_{s'} \sum_r p(s', r \mid s, a) [r + \gamma v_\pi(s')] \quad (22) \end{aligned}$$

Bellman Equation - Action-Value Function

Similarly, the **action-value function** $Q_{\pi}(s, a)$ gives the expected discounted cumulative reward from state s , taking action a and following policy π :

$$Q_{\pi}(s, a) = \mathbb{E} [R(s, a) + \gamma \mathbb{E}_{a' \sim \pi} [Q^{\pi}(s', a')] \mid s, a] . \quad (23)$$

Where R is the reward, s' the next state and a' the next action.

The **optimal** action-value function satisfies the Bellman optimality equation:

$$Q^*(s, a) = \mathbb{E} \left[R(s, a) + \gamma \max_{a'} Q^*(s', a') \mid s, a \right] . \quad (24)$$

Reinforcement Learning to Deep Reinforcement Learning

As the agent is trying to learn a value function $Q(s, a)$ or a policy function, under regular **Reinforcement Learning** it does so using a tabular approach. This approach is possible only if the state space is small and discrete, which rarely happens in a hedging scenario.

Deep reinforcement learning uses deep neural networks, which are highly flexible and ideal for approximating non-linear functions, to approximate the value function with a parameterized $Q_{\theta}(s, a)$.

Backpropagation is used to compute the required gradients to train these neural networks. Optimizers like SGD or Adam use these gradients to update the network parameters.

Neural Networks

A **Neural Network** is a group of interconnected neurons that send signals to one another, used to approximate non-linear functions.

Feed-Forward Neural Network: Neurons are organized in layers, with information flowing in one direction. For an L -layer network, we have:

$$a^{(\ell)} = \sigma \left(W^{(\ell)} a^{(\ell-1)} + b^{(\ell)} \right), \quad \ell = 1, \dots, L, \quad (25)$$

where $a^{(0)} = s$ is the input, σ is a nonlinear activation function, like ReLU, and $\theta = \{W^{(\ell)}, b^{(\ell)}\}_{\ell=1}^L$ are the trainable weights and biases.

For Deep Hedging, the output $a^{(L)}$ represents the **hedging action**, trained to approximate the optimal hedging strategy.

Training - Minibatch

Some network parameters are learned by minimizing a loss function, but it can be computationally intensive to compute the exact gradient. Two approaches are mini-batch training and Adam optimizers.

Mini-batch training: Uses small randomly sampled batches to estimate gradients, which reduces computing costs and adds stochasticity, which can improve convergence.

$$\theta \leftarrow \theta - \eta \alpha_{\theta} \frac{1}{|B|} \sum_{i \in B} \ell(\theta; \xi_i), \quad B \subset \{1, \dots, N\}, \quad |B| \ll N \quad (26)$$

Where B is the mini-batch, $|B|$ the batch size, ℓ the loss function, ξ_i the i -th training sample, and α the learning rate.

Training - Adam Optimizer

Adam optimizer: adapts the learning rate per parameter using first and second moments of the gradients:

- ▶ **First moment:** $m_t = \beta_1 m_{t-1} + (1 - \beta_1) \nabla_{\theta} L(\theta_t)$
- ▶ **Second moment:** $v_t = \beta_2 v_{t-1} + (1 - \beta_2) (\nabla_{\theta} L(\theta_t))^2$
- ▶ **Bias corrected estimates:** $\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}$
- ▶ **Weight update:** $\theta_{t+1} = \theta_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}$

where $\nabla_{\theta} L(\theta_t)$ is the gradient of the loss function with respect to the weights at time t , α is the learning rate, β_1 and β_2 are the decay rates, and ϵ is a small positive constant to avoid dividing by zero

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